A4. Monthly Attendance Announcer

Category B Assignment - Problem 4 (Intermediate)

Approach

batchPrint uses one StringBuilder, invokes displayInfo polymorphically, and uses instanceof only to guard the trainer-name downcast. It never attempts a downcast on a non-premium member.

Java Solution

```
class GymMember {
    protected final String memberId; protected final int monthlyFee;
    protected int sessionsAttended;
    public GymMember(String id,int fee) {
        if(id==null || id.trim().length()<4) throw new IllegalArgumentException("Invalid ID");
        if(fee<=0) throw new IllegalArgumentException("Fee must be positive");
        memberId=id; monthlyFee=fee;
    }
    public String displayInfo() { return "Standard | Sessions: " + sessionsAttended; }
}
class PremiumMember extends GymMember {
    private final String trainerName;
    public PremiumMember(String id,int fee,String trainer) { super(id,fee); trainerName=trainer; }
    public String getTrainerName() { return trainerName; }
    @Override public String displayInfo() {
        return "Premium | Trainer: " + trainerName + " | Sessions: " + sessionsAttended;
    }
}
class Announcement {
    static String batchPrint(GymMember[] members) {
        StringBuilder result = new StringBuilder();
        for (GymMember member : members) {
            if (result.length() > 0) result.append(" | ");
            result.append(member.displayInfo()); // virtual dispatch
            if (member instanceof PremiumMember) {
                PremiumMember premium = (PremiumMember) member;
                result.append(" [Trainer via downcast: ").append(premium.getTrainerName()).append("]");
            }
        }
        return result.toString();
    }
}
```

Sample Output

```
Standard | Sessions: 0 | Premium | Trainer: Coach Riya | Sessions: 0 [Trainer via downcast: Coach Riya] |
```